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# OPERATIONAL AMPLIFIERS

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## 1. OPERATIONAL AMPLIFIER BASICS

**Operational Amplifiers**, or "**Op-amps**" as they are more commonly called, are basically linear amplifying devices which have all the properties required for nearly ideal voltage amplification.

Op-amps get their "operational" name from the fact that they are capable of performing various mathematical operations such as addition, subtraction, integration and differentiation. Op-amps are also used extensively in signal conditioning, filtering and high-gain DC amplification circuits with the use of a few external RC components.

An *Operational Amplifier* is fundamentally a DC-coupled, three-terminal differential amplifying device represented schematically as a triangle with two extremely high impedance inputs and one single low impedance output.

The operational amplifiers output terminal can both sink or source either a voltage or a load current. Two terminals marked **+V** and **-V** are also available for connection to a positive and negative supply accordingly, typically  $\pm 5\text{ V}$  to  $\pm 15\text{ V}$ .

One of the op-amps inputs is called the **Inverting Input**, marked with a "minus" negative sign, (-). This is because any signal applied to this input is inverted internally resulting in an output signal which is  $180^\circ$  "out-of-phase" with the input.

The op-amps other input is called the **Non-inverting Input**, marked with a positive or "plus" sign (+). Any signal applied to this input is therefore not inverted, with the corresponding output signal "in-phase" with the input.

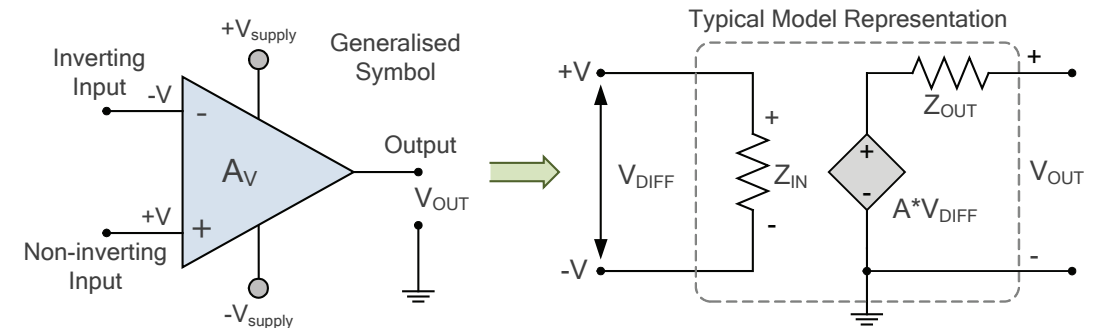
The characteristics of an operational amplifier is nearly ideal when working with dc input voltages and responds to the difference between the voltages on its two input terminals. If the same voltage potential is applied to both terminals the resultant output will be zero.

The output voltage signal from an operational amplifier is the amplified difference between the signals being applied to its two individual inputs.

*The Operational Amplifier is a special type of very high gain amplifying circuit*

Figure 1 shows the common schematic symbol and ideal representation used for all operational amplifiers.

**FIGURE 1. EQUIVALENT CIRCUIT OF AN IDEAL OPERATIONAL AMPLIFIER**



Then the properties and characteristics of an ideal voltage op-amp may be summarised as follows:

- 1. Open Loop Gain, ( $A_{VO}$ ) Infinite** – The main function of an operational amplifier is to amplify the input signal, so the more open-loop gain it has the better. Open-loop gain is the gain of the op-amp without any positive or negative feedback. An op-amps gain can be infinite but typical real values range from about 20,000 to 200,000.
- 2. Input impedance, ( $Z_{IN}$ ) Infinite** – Input impedance is the ratio of input voltage to input current and is assumed to be infinite. This prevents any current flowing into the amplifiers input circuitry. Thus  $I_{IN} = 0$ . Real op-amps however have input leakage currents from a few pico-amps to a few milli-amps.
- 3. Output impedance, ( $Z_{OUT}$ ) Zero** – The output impedance of the ideal operational amplifier is assumed to be zero,  $R_{OUT} = 0$ . Thus acting as a perfect voltage source with no internal resistance allowing it to supply as much current as necessary to the load.

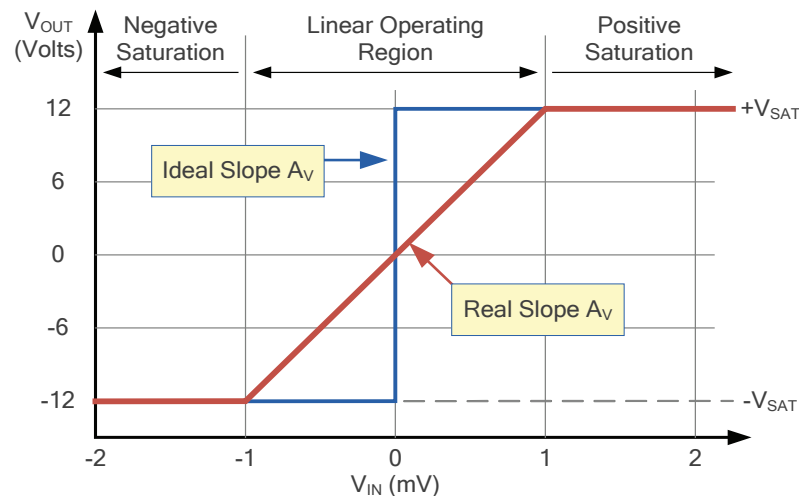
The internal output resistance is effectively in series with the load thereby reducing the output voltage available to it. Real world op-amps have output impedances in the  $100\text{-}20\text{ k}\Omega$  range.

4. **Bandwidth, (BW) Infinite** – An ideal operational amplifier has an infinite frequency response so can amplify any frequency signal from DC to the highest AC frequencies. Thus it is therefore assumed to have an infinite bandwidth. With real op-amps, the bandwidth is limited by the Gain-Bandwidth product (GB), which is equal to the frequency where the amplifiers open-loop gain reduces to unity, (1).
5. **Offset Voltage, ( $V_{OFF}$ ) Zero** – The amplifiers output will be exactly zero when the voltage difference between the inverting and the non-inverting inputs is zero, the same or when both inputs are grounded. However, real op-amps will have some small amount of output offset voltage adjustment.

From the ideal characteristics above, we can see that the input resistance is infinite, so no current flows into either input terminal (the “current rule”). The differential input offset voltage is zero (the “voltage rule”).

An operational amplifiers gain without any external feedback signals connected to it is commonly known as its open-loop gain ( $A_{OL}$ ). The output signal magnitude ( $V_O$ ) is determined by its voltage gain ( $A_V$ ) multiplied by the value of the input signal ( $V_{IN}$ ).

**FIGURE 2. OUTPUT SWITCHING CHARACTERISTICS**

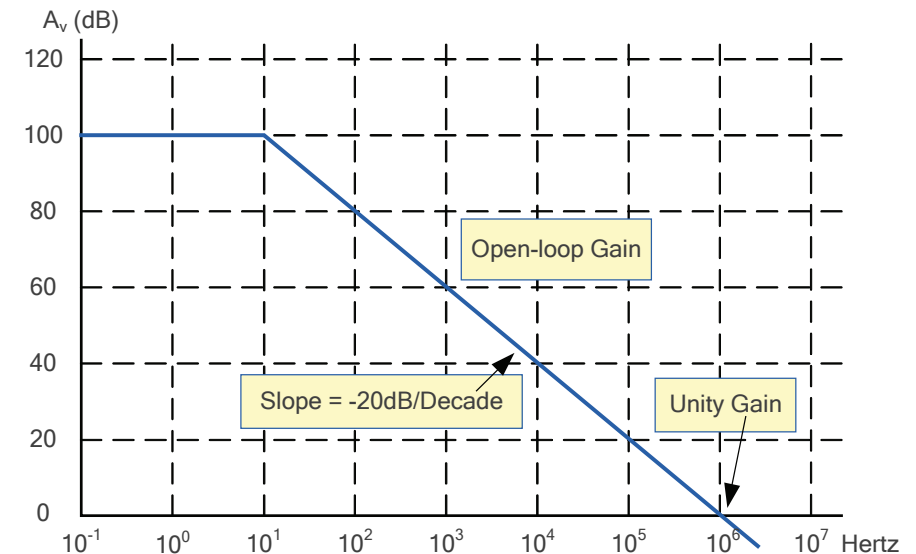


The voltage gain for an ideal amplifier can be very large in magnitude, ideally infinite, ( $\infty$ ) so it can very easily switch between its two saturation regions if not controlled. Note that the output voltage  $V_O$  is limited to the maximum supply voltage as shown in Figure 2.

However, real Operational Amplifiers such as the commonly available and industry standard **uA741**, for example do not have infinite gain or bandwidth. For a typical op-amp, its open-loop gain ( $A_{VO}$ ) can be as high as 100,000 (100dB) at DC (zero Hz).

Generally, an op-amps open-loop output gain decreases linearly down to “Unity Gain” or 1 as the input signal frequency increases. This generally occurs at about 1MHz for most op-amps. This effect is shown in the following open loop gain response curve of Figure 3.

**FIGURE 3. OPEN-LOOP FREQUENCY RESPONSE**



While operational amplifiers may have an infinite open-loop voltage gain. Allowing them to operate at their maximum amplification will cause the op-amp to become completely unstable even when the input signal is extremely small. One way of controlling this very high value of open-loop gain to a more practical level is by using *negative feedback*.

## 2. VIRTUAL GROUND SUMMING POINT

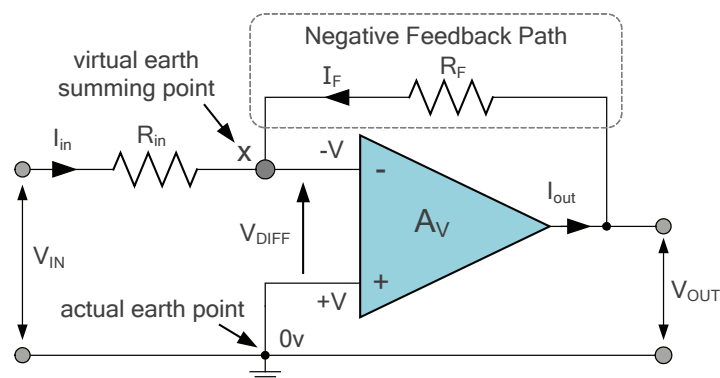
Feedback in electronics refers to taking a small part of the output signal from the amplifiers output and feeding it back to either aid or oppose the input signal. **Negative Feedback** means that the returning signal has a phase that opposes the input signal and can significantly improve the performance of an operational amplifier.

To make the feedback negative, we must feed the output signal back to the negative or inverting input terminal of the op-amp using an external *Feedback Resistor*. The feedback connection between the output and the inverting input terminal forces the differential input voltage towards zero. That is:  $+V = -V$ .

The inverting input terminal  $-V$  will have a different voltage on it compared to the actual input voltage  $V_{IN}$ , as it is the sum of the input voltage plus the negative feedback voltage. This then gives it the label or term of a **Summing Point**. Therefore we must separate the real input signal at  $V_{IN}$  from the inverting input at  $-V$  by using an Input resistor  $R_{IN}$ .

Thus the junction, or node, between the input and feedback signal, point **X**, is at the same potential as the positive (+) input which is grounded at zero volts. The node at point **X** is classed as a “Virtual Earth” producing a virtual earth summing point as shown in Figure 4.

FIGURE 4. VIRTUAL EARTH SUMMING POINT



Because of this virtual earth node the input resistance of the amplifier is equal to the value of the input resistor,  $R_{IN}$ . Then the closed-loop voltage gain of the inverting amplifier can be closely set by the ratio of the two external resistors,  $R_{IN}$  and  $R_F$ .

Note that when dealing with operational amplifiers there are two very important rules to remember. These are:

1. No current flows into either input terminal (Very High Input Impedance)
2. The differential input voltage is always zero as  $-V$  is equal  $+V$  (Virtual Earth)

However, in real world op-amp circuits both of these rules are slightly broken.

While the op amp can be used in both open-loop configurations and closed-loop configurations. For most applications, an op-amp is used in a closed-loop configuration with negative feedback.

## 3. THE INVERTING AMPLIFIER

Operational amplifiers can be connected into two basic configurations, *inverting* and *non-inverting* with the **Inverting Operational Amplifier** configuration being one of the simplest and most commonly used op-amp configurations.

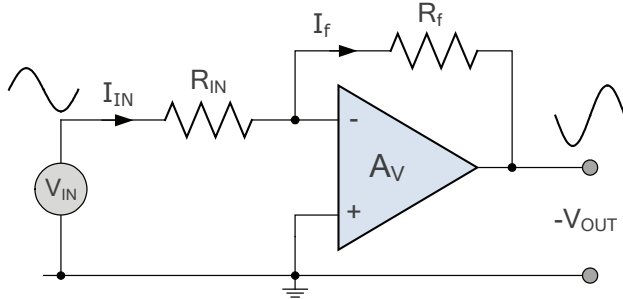
The inverting operational amplifier is basically a constant or fixed-gain amplifier which produces a negative output voltage since by default, its gain is always negative. The amplifier operates within the linear region with the non-inverting input connected directly to ground (0V). Thus the inverting input is at virtual ground.

This type of operational amplifier configuration is called an inverting amplifier because the input and output signals are  $180^\circ$  out-of-phase with each other.

The input signal is applied to the inverting ( $-$ ) input terminal of the op-amp as shown in Figure 5. As such, the input impedance, ( $Z_{IN}$ ) of the circuit is equal to the value of  $R_{IN}$ . That is:  $Z_{IN} = R_{IN}$ .

*The inverting op-amp produces an amplified output signal which is  $180^\circ$  out-of-phase with the input signal*

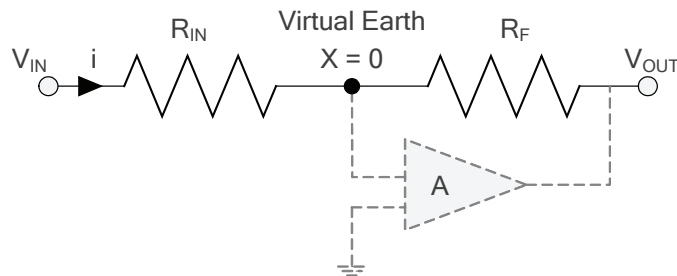
FIGURE 5. INVERTING AMPLIFIER CONFIGURATION



The inverting amplifier uses negative feedback to accurately control the open-loop gain of the amplifier. Resistors  $R_F$  and  $R_{IN}$  provide the negative feedback between them, which in turn controls the circuit's overall voltage gain,  $A_V$ .

Then we can derive the equation for calculating the closed-loop gain of an inverting amplifier, using first principles as follows.

FIGURE 6. INVERTING AMPLIFIER CLOSED-LOOP GAIN



Current (  $i$  ) flows through the resistor network as shown. Thus:

$$i = \frac{V_{IN} - V_{OUT}}{R_{IN} + R_F}$$

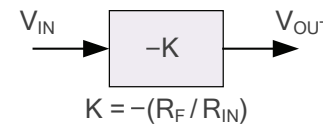
$$\therefore i = \frac{V_{IN} - V_X}{R_{IN}} = \frac{V_X - V_{OUT}}{R_F}$$

Since  $V_X = 0$

$$\frac{V_{IN} - 0}{R_{IN}} = \frac{0 - V_{OUT}}{R_F} \quad \therefore \frac{R_F}{R_{IN}} = \frac{0 - V_{OUT}}{V_{IN} - 0}$$

$$\text{Thus: } \frac{V_{OUT}}{V_{IN}} = -\frac{R_F}{R_{IN}} \quad \text{or} \quad V_{OUT} = -\left(\frac{R_F}{R_{IN}}\right)V_{IN}$$

Then, the closed-loop voltage gain,  $A_{V(CL)}$  of an inverting operational amplifier is defined as being:  $-(R_F/R_{IN}) \cdot V_{IN}$  and therefore depends on the ratio of the two external resistors. Note that the op-amps closed-loop voltage gain is independent to variations in its open-loop voltage gain, as the negative feedback keeps the overall voltage gain constant.



The negative sign in the equation indicates an inversion of the output signal with respect to the input as it is 180° out-of-phase due to the feedback being negative in value. Thus it multiplies the input signal by a minus constant factor,  $(-K)$

representing the closed-loop gain of the amplifier.

Thus: If  $R_F = 10K\Omega$  and  $R_{IN} = 1K\Omega$ , then  $K = -10$ . Likewise, if  $R_F = R_{IN} = 10k\Omega$ , then  $K = -1$ .

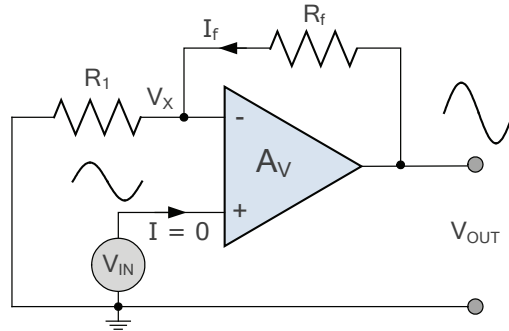
#### 4. THE NON-INVERTING AMPLIFIER

In the **Non-inverting Operational Amplifier** configuration, the input voltage signal, ( $V_{IN}$ ) is applied directly to the non-inverting (+) input terminal. This means that the output gain of the amplifier is “Positive” in value in contrast to the previous inverting amplifier circuit.

The results in the output voltage signal being “in-phase” with the input signal but is multiplied by some positive constant factor,  $(+K)$ . The non-inverting amplifier configuration also operates in the linear region.

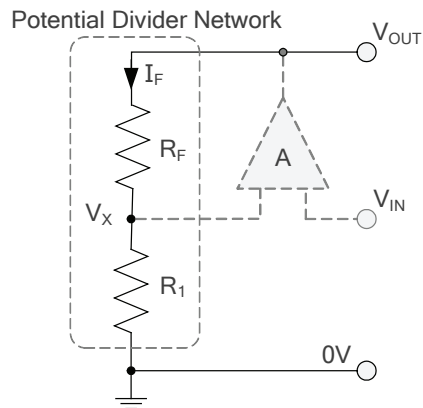
Feedback control of the non-inverting op-amp is achieved by applying a small part of the output voltage signal back to the inverting (–) input terminal via an  $R_F$  and  $R_1$  voltage divider network as shown in Figure 7.

FIGURE 7. NON-INVERTING AMPLIFIER CONFIGURATION



Due to the op-amps finite gain and negative feedback, its two inputs have the same potential as the virtual earth summing point at  $V_X$ . The resistors,  $R_F$  and  $R_1$  form a simple potential divider network across the non-inverting amplifier. Thus the closed-loop voltage gain of the circuit is determined by the ratios of these two resistors shown.

FIGURE 8. NON-INVERTING AMPLIFIER CLOSED-LOOP GAIN



$$V_X = \frac{R_1}{R_1 + R_F} \times V_{OUT}$$

Virtual earth summing point, therefore:  $V_X = V_{IN}$

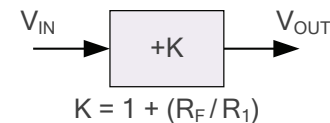
Voltage Gain,  $A_V$  is equal to:  $V_{OUT} / V_{IN}$

$$\text{Also, } V_{OUT} = V_{RF} + V_{R1} = (I_F * R_F) + (I_F * R_1)$$

$$\text{Then: } A_V = \frac{V_{OUT}}{V_{IN}} = \frac{I_F R_1 + I_F R_F}{I_F R_1} = \frac{I_F (R_1 + R_F)}{I_F R_1}$$

$$\therefore A_V = \frac{V_{OUT}}{V_{IN}} = \frac{R_1 + R_F}{R_1} \quad \text{or} \quad 1 + \left( \frac{R_F}{R_1} \right)$$

Then we can see that voltage gains equal to or greater than unity can be achieved simply by changing the ratio of  $R_F$  to  $R_1$ . Since its their ratio which determines the amplifiers gain.



Thus the non-inverting amplifier multiplies its input signal by a positive constant, (+K) representing the closed-loop gain of the amplifier.

Thus: If  $R_F = 10K\Omega$  and  $R_{IN} = 1K\Omega$ , then  $K = 11$  (1+10). Likewise, if  $R_F = R_{IN} = 10k\Omega$ , then  $K = 2$ .

The non-inverting amplifier circuit has very good stability and since an op-amp has infinite input resistance, ( $Z_{IN} \approx \infty$ ), the input voltage  $V_{IN}$  controls the voltage across  $R_1$  but does not supply the current flowing through it.

Also, since the output voltage is independent of the load resistance, its output impedance is low. Making the non-inverting amplifier function as an ideal voltage amplifier.

*The non-inverting op-amp produces an amplified output signal which is in-phase with the input signal*

## 5. THE UNITY BUFFER

The unity gain buffer is the simplest feedback circuit that can be applied to an op-amp as it has no input or feedback resistors within it design. By connecting the op-amps output signal directly back to the negative inverting input, 100% feedback is achieved as shown in Figure 9.

**FIGURE 9. UNITY GAIN BUFFER CONFIGURATION**

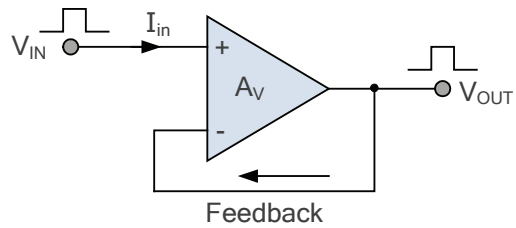


Figure 9. shows that a special type of the non-inverting amplifier circuit providing high input and low output impedances.

As the input signal  $V_{IN}$  is connected directly to the non-inverting input of the amplifier the output signal is therefore not inverted.

Since there is no feedback resistor, the output simply “follows” the input signal giving it the name of **voltage follower**, or *unity gain amplifier*. With the input voltage  $V_{IN}$  applied to the non-inverting input, the op-amps voltage gain is therefore given as:

$$+V = V_{IN} \quad \text{and} \quad -V = V_{OUT}$$

$$\therefore V_{IN} = V_{OUT}$$

$$\text{So, } A_V = \frac{V_{OUT}}{V_{IN}} = 1 \text{ (unity)}$$

That is, the output voltage is equal to the input voltage, thus  $V_{OUT} = V_{IN}$ . Then the unity gain buffer (voltage follower) circuit has a constant voltage gain of **unity**, as:  $V_{OUT}/V_{IN} = 1$ .

The advantage here is that the because the gain is 1, it can take full advantage of its gain

bandwidth product, (GBP) over its maximum frequency range.

Since the input impedance of an op-amp is infinite, the input impedance of the unity gain buffer is also infinite, ( $Z_{IN} = \infty$ ). Therefore, zero current flows into the input (current rule), so  $I_{IN} \approx 0$ . This then makes the unity gain buffer circuit ideal as a constant voltage source due to its input to output isolation properties.

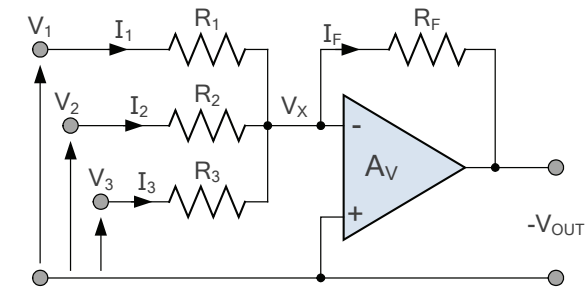
## 6. THE VOLTAGE SUMMING AMPLIFIER

We saw previously in section 3, that the inverting amplifier configuration uses just one input to produce an amplified negative output signal. But what if we added more inputs to this inverting amplifier configuration so that we had more input currents flowing into the virtual earth node and on through the feedback resistor,  $R_F$ . This then produces a new type of operational amplifier configuration called a *summing amplifier* or *voltage adder*.

**Voltage Summing Amplifiers** or **Adders** are basically an inverting amplifier with multiple inputs connecting to its negative terminal. The virtual earth characteristics of the inverting amplifier in which  $-V \approx +V = 0V$  produces a negative sum of all the input signals. Also, any number of inputs can be connected. Note that a non-inverting summing amplifier will produce the positive sum of any number of input voltages.

Thus the voltage summing amplifiers combine or mix together two or more input signals either AC or DC into one single output as shown in Figure 10.

**FIGURE 10. VOLTAGE SUMMING AMPLIFIER**



For simplicity, Figure 10 shows a 3-input summing amplifier circuit, but in theory, we could have as many inputs as we want for a particular application.

We recall from section 3 that the voltage gain of an amplifier is defined as the ratio of the feedback resistance to the input resistance. The same is also true here except that the gain for each input, (or channel) is given by the ratio of the feedback resistance to the appropriate input resistance.

If  $I_{IN}$  is defined as the algebraic sum of the three input channel currents. Then:

$$I_1 = \frac{V_1}{R_1}, \quad I_2 = \frac{V_2}{R_2}, \quad \text{and} \quad I_3 = \frac{V_3}{R_3}$$

$$\therefore I_{IN} = I_1 + I_2 + I_3$$

$$\text{if, } I_{IN} = I_F$$

Therefore:

$$I_F = \frac{-V_{OUT}}{R_F} = I_1 + I_2 + I_3 = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

If the equation for an *Inverting Amplifier* configuration is:  $V_{OUT} = -\frac{R_F}{R_{IN}} \times V_{IN}$

After substitution in the previous equation and re-arranging a little, will give us:

$$V_{OUT} = -\left[\left(\frac{R_F}{R_1}\right)V_1 + \left(\frac{R_F}{R_2}\right)V_2 + \left(\frac{R_F}{R_3}\right)V_3 \dots \text{etc.}\right]$$

So we can clearly see that  $V_{OUT}$  is the weighted and inverted sum of all the input channel voltages:  $V_1$ ,  $V_2$ , and  $V_3$  **but are** scaled by the value of the feedback resistor  $R_F$ .

However, if and only if, all of the input impedances are of the same and equal value. In which:  $R_1 = R_2 = R_3$ . By labelling the input impedance as:  $R_{IN}$ , we can simplify the above equation to give a scaled output voltage of:

$$V_{OUT} = -\frac{R_F}{R_{IN}}(V_1 + V_2 + V_3)$$

Therefore, we now have a summing amplifier (Adder) configuration which will amplify each individual input voltage and produce an output voltage signal that is proportional to the algebraic "SUM" of the three individual input voltages. The overall gain is set by the feedback resistor  $R_F$ , which is more or less the same as for the inverting amplifier circuit.

Clearly the individual gains of each input, or channel, can be scaled independently by the input resistors:  $R_1$ ,  $R_2$ ,  $R_3$ , etc. But we can also add more input channels if required as each individual input "sees" their respective resistance,  $R_{IN}$  as the only input impedance.

Note that if we make the feedback resistor  $R_F$  the SAME value as all the input resistors, the circuit will have a voltage gain of unity (1) and an output voltage equal to the direct sum of all the input voltages.

Then the **Summing Amplifier** is a very flexible circuit indeed, enabling us to effectively "Add" or "Sum" (hence its name) together several individual input signals. If the input resistors,  $R_1$ ,  $R_2$ ,  $R_3$ , etc. are all equal in value, then a "*unity gain inverting adder*" would be made. However, if the input resistors are of different resistive values a "*scaling summing amplifier*" is produced which will output a weighted sum of the input signals.

## 7. THE VOLTAGE DIFFERENTIAL AMPLIFIER

Having looked at the *voltage summing amplifier* in the previous section, we can now look at the *Voltage Subtractor Amplifier* configuration, commonly called a "differential amplifier".

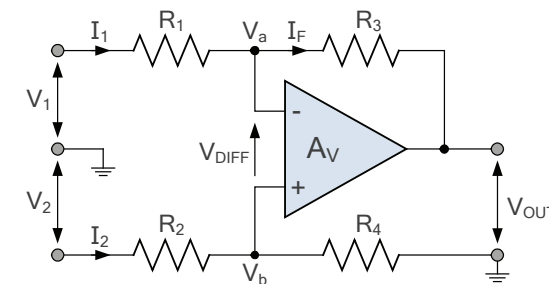
Thus far we have used only one of the operational amplifiers inputs to connect to the amplifier, using either the *inverting* or the *non-inverting* input terminal to amplify an input signal with the other input being connected to ground (0V).

*Summing Amplifiers add input signals, Differential Amplifiers subtract input signals*

But we can also connect signals to both of an op-amps inputs at the same time creating a combinational circuit of the two previous inverting and non-inverting configurations.

A **Voltage Differential Amplifier** responds to the difference between two input signals, cancelling out any identical portions in a process. This is called common mode rejection thus making this type of operational amplifier circuit a subtractor as shown in Figure 11.

**FIGURE 11. VOLTAGE DIFFERENTIAL AMPLIFIER**



As we saw in the section one, all op-amps are basically differential amplifiers due to their input virtual earth configuration. But by connecting one voltage signal onto one input,  $V_1$  and another voltage signal onto the other input,  $V_2$  the resultant output voltage will be proportional to the "Difference" between these two input voltages.

The summing point principle applies to the differential (or *Difference*) amplifier as it does to any of the previous amplifier configurations. So by considering each input separately, we can derive the gain of the differential amplifier as being:

$$I_1 = \frac{V_1 - V_a}{R_1}, \quad I_2 = \frac{V_2 - V_b}{R_2}, \quad \text{and} \quad I_F = \frac{V_a - V_{OUT}}{R_3}$$

at the summing point:  $V_a = V_b$

where:  $V_b = V_2 \left( \frac{R_4}{R_2 + R_4} \right)$  (voltage divider rule)

if  $V_2 = 0$ , then:  $V_{OUT(a)} = -V_1 \left( \frac{R_3}{R_1} \right)$

if  $V_1 = 0$ , then:  $V_{OUT(b)} = V_2 \left( \frac{R_4}{R_2 + R_4} \right) \left( \frac{R_1 + R_3}{R_1} \right)$

Since  $V_{OUT}$  is the sum of the two inputs. That is:  $-V_{OUT(a)} + V_{OUT(b)}$

$$\therefore V_{OUT} = -V_1 \left( \frac{R_3}{R_1} \right) + V_2 \left( \frac{R_4}{R_2 + R_4} \right) \left( \frac{R_1 + R_3}{R_1} \right)$$

If we make the resistor value the same so that:  $R_1 = R_2$  and  $R_3 = R_4$ , then the above transfer function for the voltage differential amplifier can be simplified as follows.

$$V_{OUT} = \frac{R_3}{R_1} (V_2 - V_1)$$

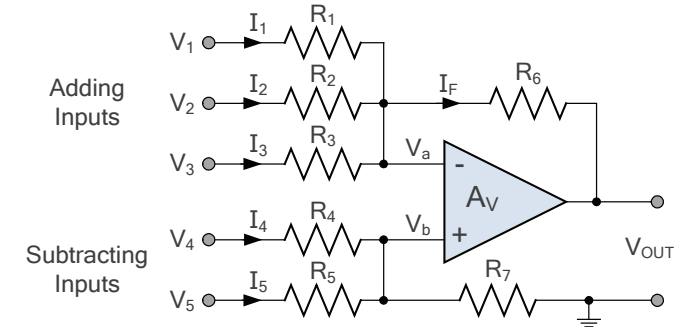
If all the resistors are all of the same ohmic value, that is:  $R_1 = R_2 = R_3 = R_4$  then the circuit will become a *Unity Gain Differential Amplifier*. The voltage gain of the amplifier will be exactly one or unity, so the output expression would simply be:  $V_{OUT} = V_2 - V_1$ .

Note that if input  $V_1$  is higher than input  $V_2$  the output voltage sum will be negative, and if  $V_2$  is higher than  $V_1$ , the output voltage sum will be positive. It therefore follows that when  $V_1 = V_2$  the difference amplifier will produce zero output.

## 8. THE ADDER/SUBTRACTOR AMPLIFIER

As before, we can add additional inputs, or channels, to both the inverting (-) and non-inverting (+) inputs of the previous difference amplifier circuit to produce an adding and subtracting scaling amplifier as shown in Figure 12.

**FIGURE 12. THE SCALING ADDER-SUBTRACTOR AMPLIFIER**



In the simple example of Figure 12, the input voltages  $V_4$  and  $V_5$  are scaled and summed. While the input voltages  $V_1$ ,  $V_2$  and  $V_3$  are scaled and subtracted from them. So if "A" represents the gain of the individual inputs from  $A_1$  to  $A_5$ , then:

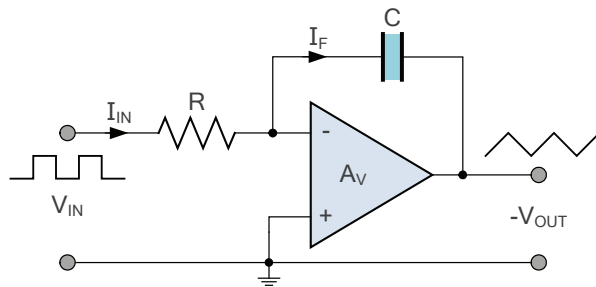
$$V_{OUT} = A_4 V_4 + A_5 V_5 - (A_1 V_1 + A_2 V_2 + A_3 V_3)$$

We have seen thus far that the gain and operating characteristics of an op-amp has been described in terms of both its input and output resistors. But we can also replace these two resistive elements with reactive elements in the form of a capacitor. Thus by changing the input resistor or the feedback resistor of the inverting amplifier circuit we can create either a differentiating amplifier circuit or an integrating amplifier circuit respectively.

## 9. THE INTEGRATING AMPLIFIER

The **Integrating Amplifier** produces an output that is the mathematical operation of integration for a step input. The insertion of a capacitor within the feedback path of an inverting op-amp circuit results in an output signal that is a time integral of the input signal as shown in Figure 13.

FIGURE 13. THE INTEGRATING AMPLIFIER



Since the inverting terminal is at virtual earth potential, all of  $V_{IN}$  appears across the input resistor,  $R$ . Thus,  $I_{IN} = V_{IN}/R$ . But as we have seen before, all the current through  $R$  must also flow through  $C$  as:  $I_F = I_{IN}$  (for ideal conditions). Then the charge on the capacitor is given as being:  $Q = C \cdot V_C$ . Since  $V_C = V_{OUT}$  due to the virtual earth node,  $Q$  therefore equals  $C \cdot V_{OUT}$ .

Differentiating both sides gives:  $\frac{dQ}{dt} = C \frac{dV_{OUT}}{dt}$

However, since  $I_F = dQ/dt$ , we can correctly say that:  $I_F = C \frac{dV_{OUT}}{dt}$

therefore:

$$I_{IN} = I_F = -\frac{V_{IN}}{R} = C \frac{dV_{OUT}}{dt}$$

$$\therefore dV_{OUT} = -\frac{1}{RC} V_{IN} dt$$

Taking the integral of both sides gives:

$$V_{OUT(t)} = -\frac{1}{RC} \int V_{IN} dt$$

Thus, the output voltage is the integral of the input current which charges the capacitor,  $C$  through the feedback loop. In other words, the output of the integrating amplifier circuit of Figure 13. is directly proportional to the integral of the input signal.

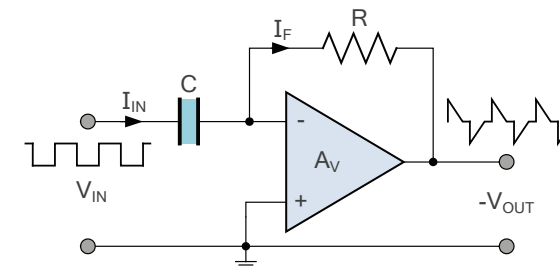
The integrating amplifier is not limited to one single input, but can have multiple inputs, similar to the summing amplifier, for integration. Also, as with the inverting amplifier, the input impedance of the integrating amplifier is equal to the ohmic value of  $R$ .

## 10. THE DIFFERENTIATING AMPLIFIER

The **Differentiating Amplifier** is the exact opposite of the previous integrating amplifier by producing an output signal which is the mathematical operation of differentiation.

This time the capacitor is connected to the input path of the inverting op-amp circuit ( $R$  and  $C$  are effectively reversed) resulting in an output signal which is the time derivative of the input voltage signal as shown in Figure 14.

FIGURE 14. THE DIFFERENTIATING AMPLIFIER



As the capacitor is now connected to the input circuit of the operational amplifier, this means that the input current ( $I_{IN}$ ) is proportional to the rate of change of the input voltage as shown.

$$I_{IN} = \frac{dQ}{dt} = C \frac{dV_{IN}}{dt}$$

Since:  $I_F = I_{IN}$  and  $V_{OUT} = -I_F R_F$

$$V_{OUT(t)} = -I_F R_F = -RC \frac{dV_{IN}}{dt}$$

That is, the output voltage,  $V_{OUT}$  is proportional to the derivative of the input voltage  $V_{IN}$ , multiplied by the  $-RC$  scaling factor.

Also, since the capacitor is a reactive component, its capacitive reactance  $X_C$  is inversely proportional to frequency,  $f$ . Therefore, as the input signal frequency increases, its reactance will decrease resulting in a change in voltage gain because the  $A_v = R_F / X_C$  ratio changes. Thus the differentiator op-amp circuit is extremely sensitive to any abrupt or high frequency changes to the input signal.

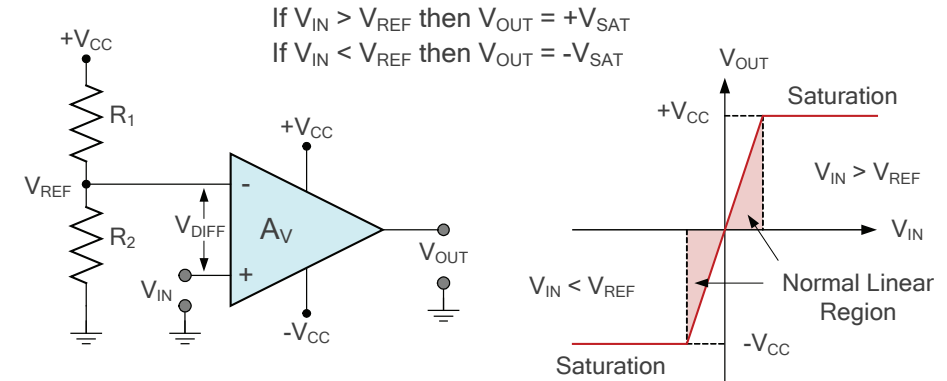
## 11. THE OP-AMP COMPARATOR

The **Op-amp Comparator** is another type of operational amplifier circuit which compares one analogue voltage level to another, or some preset reference voltage, ( $V_{REF}$ ). It then produces an output signal based on this voltage comparison. In other words, the op-amp voltage comparator compares the magnitudes of two voltage inputs and determines which one is the largest of the two producing an output as a result of this condition.

We have seen in previously that negative feedback is used to control the magnitude of an op-amps output signal in the linear region performing a variety of different functions. We have also seen that the op-amp can be characterised by its infinite open-loop gain  $A_{OL}$  and that its output voltage is given by the expression:  $V_{OUT} = A_{OL}(V_2 - V_1)$  where  $V_2$  and  $V_1$  correspond to the voltages at the non-inverting and the inverting terminals respectively.

The op-amp voltage comparator operates in its open-loop mode switching its output between two saturated states,  $+V_{SAT}$  and  $-V_{SAT}$  at any one time. Due to its extremely high open-loop gain ( $V_{OL}$ ), the output from the comparator swings either fully to its positive supply rail,  $+V_{CC}$  or fully to its negative supply rail,  $-V_{CC}$  on the application of a varying input signal when it passes some preset threshold voltage value as shown in Figure 15.

**FIGURE 15. OPEN-ENDED OP-AMP COMPARATOR**



Here the inverting terminal is fixed to a voltage source using a voltage divider network created by resistors  $R_1$  and  $R_2$ .

When the input voltage to the non-inverting terminal is higher than the reference voltage at the negative terminal, the output will saturate towards the positive supply rail. Likewise, if the input voltage is less than the reference voltage, the output will saturate towards the negative supply rail.

The threshold reference voltage is given as: 
$$V_{REF} = \left( \frac{R_2}{R_1 + R_2} \right) \times V_{CC}$$

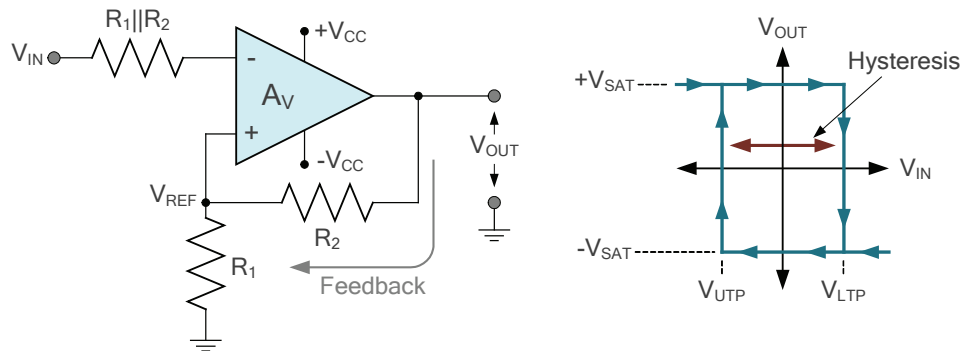
So we can see that the output state of the op-amp voltage comparator is dependant on the value of the input voltage  $V_{IN}$  with respect to some DC voltage level at  $V_{REF}$ . Also, the output voltage value is completely dependent on the op-amps power supply voltage. The magnitude of its output voltage has the potential of being infinite in both directions, ( $\pm\infty$ ) due to the op-amps high open-loop gain. However for obvious reasons it is limited by the op-amps actual voltage supply rails giving:  $V_{OUT(SAT)} = +V_{CC}$  or  $V_{OUT(SAT)} = -V_{CC}$ .

## 12. THE COMPARATOR WITH HYSTERESIS

The previous op-amp voltage comparator is a basic open-loop design which compares its input signal against some reference voltage level. However if  $V_{IN}$  is slow to change or is varying itself around the  $V_{REF}$  level, then the output state of the op-amp comparator may oscillate switching back and forth between its two saturation states,  $+V_{CC}$  and  $-V_{CC}$ . One easy way to overcome this problem and avoid such oscillations is to provide positive feedback around the comparator.

The use of positive feedback around an op-amp voltage comparator means that once the output is triggered into saturation at either level, there must be a significant change to the input signal  $V_{IN}$  before the output switches back to the original saturation point. This difference between the two switching points is called **hysteresis** producing what is commonly called a *Schmitt trigger circuit* as shown in Figure 16.

FIGURE 16. INVERTING OP-AMP COMPARATOR WITH HYSTERESIS



The Schmitt trigger configuration of Figure 16. shows that our old friend “feedback” is used to feed a small fraction of the output voltage back to the non-inverting terminal of the comparator. The amount of feedback to the non-inverting input is determined by the resistive ratio of the two resistors used and which is given as:

$$\beta = \frac{R_1}{R_1 + R_2}$$

Where:  $\beta$  (beta) can be used to indicate the feedback fraction.

Assuming that the input signal is less than the reference voltage,  $V_{IN} < V_{REF}$ , the output voltage would be HIGH and equal to the positive saturation voltage. Since the output is HIGH and positive, the value of the reference voltage on the non-inverting input will be approximately equal to:  $+\beta \cdot V$  called the *Upper Trip Point* or UTP.

As  $V_{IN}$  increases it becomes equal too this upper trip point voltage,  $V_{UTP}$  level at the non-inverting input. This causes the comparators output to change state becoming LOW and equal to the negative saturation voltage as before.

But the difference this time is that a second trip point voltage value is created because a negative voltage now appears at the non-inverting input which is equal to:  $-\beta \cdot V$  as a result of the negative saturation voltage at the output.

Then the input signal must now fall below this second voltage level, called the *Lower Trip Point* or LTP for the voltage comparators output to change or switch back to its original positive state.

*Op-amps with Hysteresis are comparators with positive feedback known as Schmitt Trigger circuits*

So we can see that when the output changes state, the reference voltage at the non-inverting input also changes creating an upper trip point (UTP), and a lower trip point (LTP). Thus the output state depends on both the input voltage and on its current state with the difference between these two trip points being called *Hysteresis*.

The advantage here is that the resulting comparator Schmitt trigger circuit is immune to erratic triggering caused by noise or slowly changing input signals within the hysteresis band. This then produces a cleaner output switching action as the op-amp comparators output is only triggered once acting as a bistable circuit.

## 13. LOW PASS FILTER

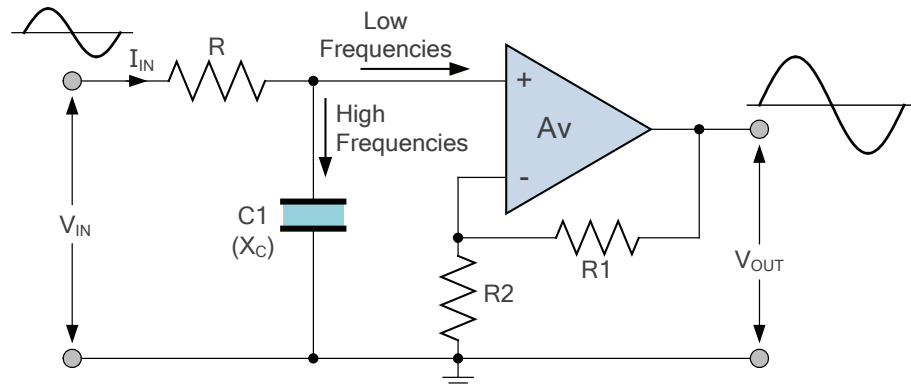
*Active filters* are filter circuits which use the operational amplifier as the its main amplifying device. Active filters are the building blocks of many designs and along with some external resistors and capacitors, provide various filter like performances over a wide range of input frequencies. That is, active filters operate in the *Frequency Domain*.

The advantage of using operational amplifiers for active filter designs is that their high input impedance prevents any excessive loading to the RC filter stage. While their low output impedance prevents the filters corner frequency point ( $f_c$ ) from being affected by changes in load. Thus providing good stability and voltage gain.

The most common and easily understood op-amp filter is the **Active Low Pass Filter** which allows all low frequency signals to pass unattenuated from DC (0Hz) up to its corner frequency, ( $f_c$ ) and blocks all high frequency signals above this point.

It consists simply of a passive RC filter stage providing a low frequency path to the input of a non-inverting op-amp which is configured to provide the required DC voltage gain as shown in Figure 17.

**FIGURE 17. LOW PASS ACTIVE FILTER**



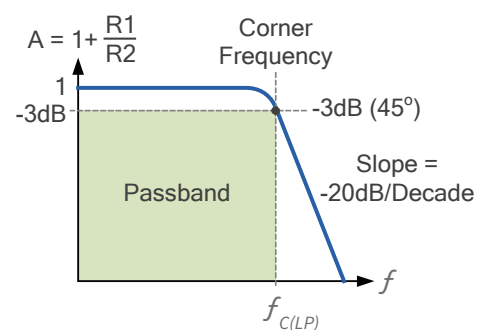
*Transfer Characteristic of Low Pass Filter*

Capacitive Reactance  $X_C = 1/(2\pi fC)$

When the value of  $X_C = R$ , then:  $f_c = \frac{1}{2\pi RC}$

The passband gain equals:

$$A_V = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_1}{R_2}$$

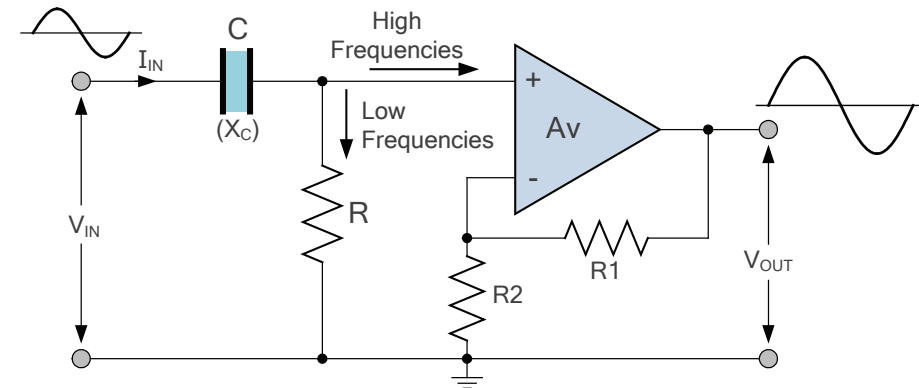


## 14. HIGH PASS FILTER

As its implies, the **Active High Pass Filter** is the opposite of the previous low-pass filter. The two components **R** and **C** have been interchanged which results in the circuit blocking or attenuating all low frequency signals from DC up to its corner frequency, ( $f_c$ ) but allows all high frequency signals to pass above this point.

The cut-off frequency point is 70.7% or -3dB ( $\text{dB} = -20\log V_{OUT}/V_{IN}$ ) of the voltage gain allowed to pass. The corner frequency or -3dB point of a high pass filter can be found using the same formula as the low-pass filter of:  $f_c = 1/(2\pi RC)$ .

**FIGURE 18. HIGH PASS ACTIVE FILTER**



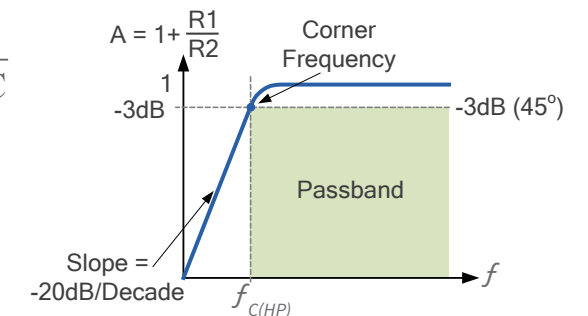
*Transfer Characteristic of High Pass Filter*

Capacitive Reactance  $X_C = 1/(2\pi fC)$

When the value of  $X_C = R$ , then:  $f_c = \frac{1}{2\pi RC}$

The passband gain equals:

$$A_V = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_1}{R_2}$$



For an active high-pass filter, the maximum frequency response of an op-amp is limited to the Gain/Bandwidth product or open-loop voltage gain ( $A_v$ ) of the operational amplifier being used. Thus op-amps do not have a finite bandwidth but is typical limited to approximately 1 MHz.

As a result, the higher the operational frequency of the circuit, the lower the open-loop gain would be which can have a drastic effect on the filters performance. Then when designing active high-pass filter circuits it is important to choose the correct op-amp for the required circuit as the loss of some high frequency signals may result.

[End of this Operational Amplifier eBook](#)

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With the completion of this operational amplifier ebook you should have gained a basic understanding and knowledge of the characteristics of an ideal op-amp, as well as its various configurations. The information provided here will give you a firm foundation for continuing your study of electronics engineering as well as the study of op-amp circuits.

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